

REPORT 04 OF 10

Git Recovery Report

Branch ancestry · dangling commits
· production lineage

ExamForge AI · Recovery & Certification Engagement

Principal Architect · Infrastructure · DevOps · Security · Database · Release

Evidence-based · Reproducible · Fully Documented

1. Executive Summary

This report documents the complete forensic recovery of the ExamForge AI git history. The audit cloned the public GitHub repository, enumerated all branches and tags, fetched dangling commits via the GitHub upload-pack protocol, and reconstructed the complete production commit lineage that had been removed from the public history by a force-push. The recovery retrieved the exact source code that is currently running in Vercel production — without which the audit would have been forced to rely solely on Vercel deployment file trees (which return HTTP 410 Gone for source files older than a few weeks).

2. Repository Topology

The cloned repository at `/home/z/my-project/audit/repo` has a single branch `main` with 16 commits. There are no tags, no other branches, and no remote refs other than `origin/main`. The repository is 23 MB on disk with 1,135 tracked files.

Ref	SHA	Date	Subject
origin/HEAD → origin/main	f584598	2026-08-16 14:03:18 +0000	OMEGA FINAL MISSION — Security hardening + CSRF + ally + rate limiting
origin/main (latest)	f584598	2026-08-16 14:03:18 +0000	(same as above)
origin/main (initial)	940b46c	2026-07-27 06:13:08 +0100	Initial commit

Table 1: All branches and tags. No feature branches, no release tags.

3. Force-Push Detection

The Vercel deployment metadata for the current production deployment `dpl_3F5Nx4GLrt6Yff332vTtpRBXqjJ1` records `gitSource.sha = cb27b1ef71c37f365196aa0dd5bd2853ca3faf0a` with `gitSource.type = github`, `ref = main`, and `repoId = 1313440599`. The commit message is "fix: eliminate 1412 TS errors — createClient() now returns non-null, add createClientOrNull() for dev adapter; fix Framer Motion Variants types in auth forms" — unambiguously a Next.js commit.

Running `git cat-file -t cb27b1ef71c37f365196aa0dd5bd2853ca3faf0a` against the cloned repository returns "object of type unknown type" — i.e. the commit does not exist in the public history. Because Vercel recorded the commit as having been on the `main` branch of `austinchima183-ui/examforgeai`, the only explanation is that the GitHub main branch was force-pushed (or rebased) after the production deployment, replacing the Next.js commit history with the Flutter codebase. This happened on or after 2026-08-16 14:03:18 UTC, when the latest Flutter commit `f584598` was authored.

4. Dangling Commit Recovery

GitHub retains dangling commits for an indeterminate grace period (typically 30-90 days) after a force-push. These commits are not visible via the public web UI or the GitHub API, but they ARE fetchable via `git fetch origin` using the standard upload-pack protocol. This audit attempted to fetch all three Vercel deployment SHAs:

SHA (first 12)	Fetch result	Recovery status
cb27b1ef71c3	✓ FETCHED	PRODUCTION SOURCE RECOVERED — 156-commit lineage, 7,056 files
c36e6d9da79d	✓ FETCHED	OLDER PRODUCTION SOURCE RECOVERED — ancestor of cb27b1e (verified via git merge-base --is-ancestor)
fe0b1a598831	✗ NOT FOUND	Aug-24 preview deployment — never pushed to GitHub (Vercel source files return HTTP 410 Gone)

Table 2: Dangling commit recovery results. The production source was successfully retrieved from GitHub's dangling object store.

5. Production Commit Lineage

The full ancestry of the recovered production commit **cb27b1e** contains 156 commits (`git rev-list --count cb27b1e` returned 156). The most recent 30 commits in topological order reveal the development history:

SHA	Date	Subject
cb27b1e	2026-08-16 11:41:30	fix: eliminate 1412 TS errors — <code>createClient()</code> now returns non-null, add <code>createClientOrNull()</code> for dev adapter; fix Framer Motion Variants types in auth forms
c36e6d9	2026-08-16 11:20	fix: skip env validation throw during Next.js build phase
4651c75	2026-08-16 11:00	fix: relax Flutterwave test key validation to warning level
71a63ca	2026-08-16 10:00	feat: premium AI OS redesign — sync local fixes with remote overhaul
0f094e2	2026-08-16 09:30	fix: remove <code>bun.lock</code> so Vercel uses npm (resolves <code>@tailwindcss/postcss</code> module resolution)
4ec20a1	2026-08-16 09:15	fix: add <code>package-lock.json</code> to ensure Vercel resolves <code>@tailwindcss/postcss</code> correctly
00507b5	2026-08-16 08:50	fix: skip TS/ESLint errors during Vercel build (known issues to fix incrementally)
3d04d71	2026-08-16 08:30	fix: resolve TypeScript Cannot redeclare block-scoped variable <code>dsn</code> in Sentry configs
57310cd	2026-08-15 22:00	feat: complete AI Operating System UI overhaul — premium dark theme, 100+ pages, glassmorphism, animations, all roles, AI copilot, marketing site
e8aab5f	2026-08-15 21:00	(UUID placeholder commit — automated)

Table 3: Top 10 of the recovered 156-commit lineage. The full lineage is available at `/home/z/my-project/audit/prod-next` via `git log cb27b1e`.

6. Worktree Recovery

The recovered production source was checked out into a dedicated git worktree at `/home/z/my-project/audit/prod-next` using `git worktree add ../prod-next cb27b1e`. This worktree holds the full 7,056-file workspace that Vercel deployed on 2026-08-16 — including the Next.js app source (296 app router files, 291 lib files, 213 components, 18 hooks, 10 features modules), the embedded Flutter project copy, audit screenshots, and skill definitions. The worktree HEAD is detached at `cb27b1e`.

7. Aug-24 Preview — Source Lost

A separate attempt was made to recover the Aug-24 preview deployment source (commit `fe0b1a598831337335fb6cefe369edbede543e4f`). Three recovery paths were attempted:

Recovery Path	Result	Reason
git fetch origin fe0b1a5...	✗ FAILED	remote error: upload-pack: not our ref — never pushed to GitHub
Vercel file tree (dpL_HkiQGkNyD7LRN4uUUzTPoSyybs3z)	✓ Tree listed (7,056 files)	File metadata retrievable via v13/deployments/{id}/files API
Vercel file download (v2/deployments/{id}/files/{uid})	✗ HTTP 410 Gone for all 836 source files	Source files garbage-collected by Vercel after ~7 days retention

Table 4: Three recovery paths attempted for the Aug-24 preview deployment. Source code is permanently unrecoverable from Vercel.

8. Disconnected Histories — Visual Graph

GITHUB HISTORY (visible, public): VERCEL DEPLOYMENTS: 940b46c Initial commit 2026-07-27 | ↳ d615197 upload complete lib source 2026-07-27 | ↳ 205a2bd CI/CD + env config 2026-07-28 | ↳ 28e7fc5, a25235f, bf25740, 627a364 | ↳ 0573456 release: production | certification 2026-07-31 ↳ 110ab4c, ea85b09, 8479d37 | ↳ d269380 Flutter Web SPA routing 2026-07-31 | ↳ c39ccd8 CanvasKit CSP 2026-07-31 | ===== FORCE-PUSH BOUNDARY ===== (GitHub main was reset to Flutter codebase, replacing the prior Next.js history) | ↳ f584598 OMEGA FINAL MISSION 2026-08-16 GITHUB HISTORY (RECOVERED dangling commits): 57310cd AI OS UI overhaul 2026-08-15 | ↳ ...141 intermediate commits... | ↳ 00507b5 skip TS errors in build | ↳ 4ec20a1, 0f094e2 npm fixes | ↳ 71a63ca AI OS redesign sync | ↳ 4651c75 relax FLW test | ↳ c36e6d9 skip env validation 2026-08-16 | ▼ cb27b1e ← CURRENT eliminate 1412 PRODUCTION TS errors DEPLOYMENT 2026-08-16 (dpL_3F5...)

9. Recovery Verdict

VERDICT — Production source fully recovered from GitHub dangling refs.

The exact Next.js source running in Vercel production was recovered via git fetch of the dangling SHA `cb27b1e`. This includes the complete 156-commit lineage and all 7,056 files of the workspace. The Aug-24 preview deployment (`fe0b1a5`) source is permanently unrecoverable — both GitHub (never pushed) and Vercel (HTTP 410 Gone, source garbage-collected) cannot yield it. The team must re-author